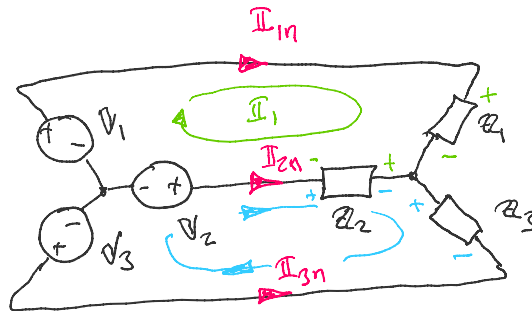


7.65)



Bestäm strömmen

 I_{1n} genom

a) MESHANALYS

b) NODANALYS

Det väner att $V_1 = 170 \angle 0^\circ$, $V_2 = 170 \angle 120^\circ$, $V_3 = 170 \angle -120^\circ$
 $Z_1 = 0.5 \angle 20^\circ \Omega$, $Z_2 = 0.35 \angle 0^\circ$, $Z_3 = 1.7 \angle -90^\circ \Omega$

Lösning: Använd MESHSTRÖMMARNA I_1 och I_2 .
 Använd KVL i resp. MESH.

$$\begin{cases} -V_1 + Z_1 I_1 + Z_2 I_1 - Z_2 I_2 + V_2 = 0 & (1) \\ -V_2 + Z_2 I_2 - Z_2 I_1 + Z_3 I_2 + V_3 = 0 & (2) \end{cases}$$

STUVA OM LIGG (1) och (2)

$$\begin{cases} V_2 - V_1 + (Z_1 + Z_2) I_1 - Z_2 I_2 = 0 & (3) \end{cases}$$

$$\begin{cases} V_3 - V_2 + (Z_2 + Z_3) I_2 - Z_2 I_1 = 0 & (4) \end{cases}$$

MEN

$$\begin{aligned} V_2 - V_1 &= 170 (\angle 120^\circ - \angle 0^\circ) = 170 (-0.5 + j\frac{\sqrt{3}}{2} - 1) = \\ &= -255 + j147.2 = \underline{294.5 \angle 150^\circ} \quad \checkmark \end{aligned} \quad (5)$$

$$\begin{aligned} V_3 - V_2 &= 170 (\angle -120^\circ - \angle 120^\circ) = 170 (-0.5 - j\frac{\sqrt{3}}{2} + 0.5 - j\frac{\sqrt{3}}{2}) \\ &= 170 \sqrt{3} \angle -90^\circ = \underline{294.5 \angle -90^\circ} \quad \checkmark \end{aligned} \quad (6)$$

och

$$\begin{aligned} Z_1 + Z_2 &= 0.5 \angle 20^\circ + 0.35 \angle 0^\circ = 0.47 + j0.17 + 0.35 = \\ &= 0.82 + j0.17 = \underline{0.84 \angle 11.71^\circ} \Omega \end{aligned} \quad (7)$$

$$\begin{aligned} Z_2 + Z_3 &= 0.35 \angle 0^\circ + 1.7 \angle -90^\circ = 0.35 - j1.7 = \\ &= \underline{1.74 \angle -78.34^\circ} \Omega \end{aligned} \quad (8)$$

(3) ger

$$I_2 = \frac{1}{Z_2} [(V_2 - V_1) + (Z_1 + Z_2) I_1] \quad (9)$$

(9) i (4) ger

$$[0.82 + j0.17] [(V_2 - V_1) + (Z_1 + Z_2) I_1] = Z_2 I_2$$

(7) i (4) gen

$$(V_3 - V_2) + \frac{R_2 + R_3}{R_2} [(V_1 - V_2) + (R_1 + R_2) I_1] = R_2 I_1$$

$$(V_3 - V_2) + \frac{R_2 + R_3}{R_2} (V_1 - V_2) = R_2 I_1 - \frac{(R_2 + R_3)(R_1 + R_2)}{R_2} I_1$$

Använd (5) - (8)

$$294.5 \angle -90^\circ + \frac{1.74 \angle -78.37^\circ}{0.35 \angle 0^\circ} 294.5 \angle 15^\circ = \left[0.35 \angle 0^\circ - \frac{1.74 \angle -78.37^\circ \cdot 0.84 \angle 111.71^\circ}{0.35 \angle 0^\circ} \right] I_1$$

$$\text{H.L.} = \left[0.35 \angle 0^\circ - \frac{1.46 \angle -66.66^\circ}{0.35 \angle 0^\circ} \right] I_1$$

$$= \left[0.35 \angle 0^\circ - 4.17 \angle -66.66^\circ \right] I_1$$

$$= \left[-1.30 + j3.83 \right] I_1 = \underline{4.05 \angle 108.75^\circ} I_1 \quad (10)$$

V.L.

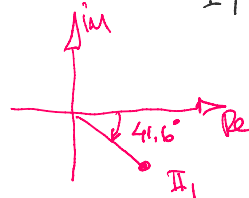
$$-j 294.5 + 1464.07 \angle 71.63^\circ = 461.41 + j1094.78 =$$

$$\underline{1188.22 \angle 67.15^\circ} \quad (11)$$

(10) & (11) gen skrivna

$$I_1 \cdot 4.05 \angle 108.75^\circ = 1188.22 \angle 67.15^\circ$$

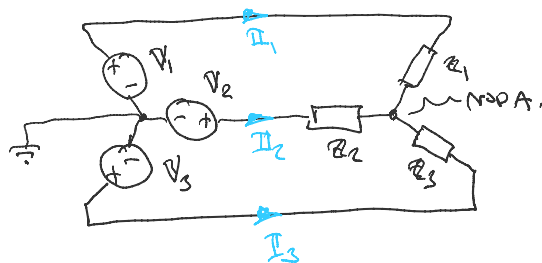
$$I_1 = \frac{1188.22 \angle 67.15^\circ}{4.05 \angle 108.75^\circ} = \underline{293.38 \angle -41.6^\circ} \text{ A.}$$



SI DÄR
UNGEFÄR

I DETTA FÄR MÖJLIGAST
MESHSTRÖMMEN I_1 , DEN
VARIERA STRÖMMEN I_{11}

b) Samma igen men med voltmätare



KSL för nod A gen (positiv till noden)

$$-I_1 - I_2 - I_3 = 0$$

men

$$\frac{V_1 - V_A}{R_1} = I_1 \quad (*)$$

$$\frac{V_2 - V_A}{R_2} = I_2$$

$$\frac{V_3 - V_A}{Z_3} = I_3$$

$$\Rightarrow \frac{V_A - V_1}{Z_1} + \frac{V_A - V_2}{Z_2} + \frac{V_A - V_3}{Z_3} = 0$$

$$V_A \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} \right) = \frac{V_1}{Z_1} + \frac{V_2}{Z_2} + \frac{V_3}{Z_3}$$

$$V_A = \frac{\frac{V_1}{Z_1} + \frac{V_2}{Z_2} + \frac{V_3}{Z_3}}{\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}}$$

Numeriskt

$$V_A = \frac{\frac{170 \angle 0^\circ}{0.5 \angle 25^\circ} + \frac{170 \angle 120^\circ}{0.35 \angle 5^\circ} + \frac{170 \angle -120^\circ}{1.7 \angle -90^\circ}}{\frac{1}{0.5 \angle 25^\circ} + \frac{1}{0.35 \angle 5^\circ} + \frac{1}{1.7 \angle -90^\circ}} =$$

$$= \frac{340 \angle -20^\circ + 485.4 \angle 120^\circ + 100 \angle -30^\circ}{2 \angle -20^\circ + 2.86 \angle 5^\circ + 0.59 \angle 90^\circ} =$$

$$= \frac{163.2 + j254.3}{4.73 - j0.094} = \frac{302.16 \angle 58.31^\circ}{4.75 \angle -1.14^\circ} =$$

$$= \underline{\underline{63.88 \angle 58.45^\circ}}$$

Strömmen I_1 blir (VR (*))

$$I_1 = \frac{V_1 - V_A}{Z_1} = \frac{170 \angle 0^\circ - 63.88 \angle 58.45^\circ}{0.5 \angle 25^\circ} =$$

$$= \frac{136.54 - j54.44}{0.5 \angle 25^\circ} = \frac{147.02 \angle -21.73^\circ}{0.5 \angle 25^\circ} =$$

$$= \underline{\underline{294.04 \angle -41.73^\circ \text{ A}}}$$

↑
 samma svar som för
 MESHANALYS (ATT DET ÄR
 LITET SKILLNAD BEROR PÅ AVRUNDNING
 I DE OLIKA BERÄKNINGSSTEGEN.)

Svar: a) $I_1 = 293 \angle -42^\circ$
 b) $I_1 = 294 \angle -42^\circ$